

PS 26176 · Team BitByBits

ORCA X MitraAI

Collaborative agentic AI marine intelligence platform

Marine decisions are data rich but intelligence poor.

A fisherman deciding whether to sail, an officer deciding which harbours to close, and a researcher assembling a dataset all need the same ocean and all of them currently get it from different portals, in different formats, at different times. ORCA is one reasoning layer over those sources that answers in plain language, shows its working, and refuses to guess when it does not know.

60

harbours scored
live

11

rescue centres
mapped

11

coastal languages

539

automated tests

What is inside?

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1. The problem, and the shape of the answer

Every number a mariner needs already exists. Wave height comes from one model, wind from another, satellite sea-surface temperature from a third, official warnings from a fourth, and maritime boundaries from a fifth. Each lives behind its own portal, its own file format and its own update clock. The work of reconciling them falls on the person least equipped to do it usually at four in the morning, on a phone, before deciding whether to put to sea.

ORCA does that reconciliation. It is a reasoning layer, not another dashboard: you ask a question in your own language, and it plans which sources to consult, fetches them, cross-checks them, and answers with a verdict plus every number behind it.

The one architectural rule everything else follows

A language model never decides whether it is safe to sail. It parses the question and writes the explanation; a deterministic rule engine produces the verdict. This is enforced in code an automated test fails the build if the decision core ever imports the agent layer.

What was missing before?

Existing approach	What it cannot do	What ORCA adds
Data portals and dashboards	Answer a question	Natural-language intent, decomposed into sub-queries
Individual provider APIs	Agree with each other	Cross-source evidence fusion with disagreement reported
Layer and map viewers	Know who is asking	Vessel-class-aware, context-aware reasoning
Static forecast bulletins	Adapt to one boat	Multi-model risk inference against cited thresholds

2. System architecture

ORCA is four layers. Data adapters fetch and normalise; a deterministic science and decision core computes verdicts; an agent layer plans and explains; a client renders. Information flows upward only the core never asks the agent layer anything, which is what makes the safety guarantee checkable rather than aspirational.

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Graceful degradation is a design property, not an accident

Every optional component has a declared fallback, and the running system reports which path it took. Postgres with PostGIS falls back to SQLite; Redis with an ARQ queue falls back to an in-process TTL cache with APScheduler; four LLM providers fall back to one another in order. The health endpoint names the active path and the reason for any fallback, so a degraded deployment is visible rather than silent.

3. The decision core: how a GO / NO-GO is computed

The verdict has two independent parts, and they are never merged. First, hard vetoes: any single limit exceeded for the vessel class produces NO-GO regardless of how good everything else looks. Second, a weighted 0-100 comfort index for the cases where nothing is vetoed.

$$\text{Index} = 0.35 \cdot S_{\text{wave}} + 0.30 \cdot S_{\text{wind}} + 0.15 \cdot S_{\text{visibility}} + 0.20 \cdot S_{\text{instability}}$$

$$S_{\text{wave}} = \max(0, 100 \cdot (1 - (H_s / H_{s_limit})^2))$$

$$S_{\text{wind}} = \max(0, 100 \cdot (1 - (U_{10} / U_{limit})^2))$$

$$S_{\text{vis}} = \min(100, (V / 10 \text{ km}) \cdot 100)$$

Wave carries the largest weight because it is what capsizes a small hull. Both wave and wind terms are quadratic: a 2 m sea is far more than twice as dangerous as a 1 m sea to an 8 m boat, and a linear score would understate exactly the region where boats are lost.

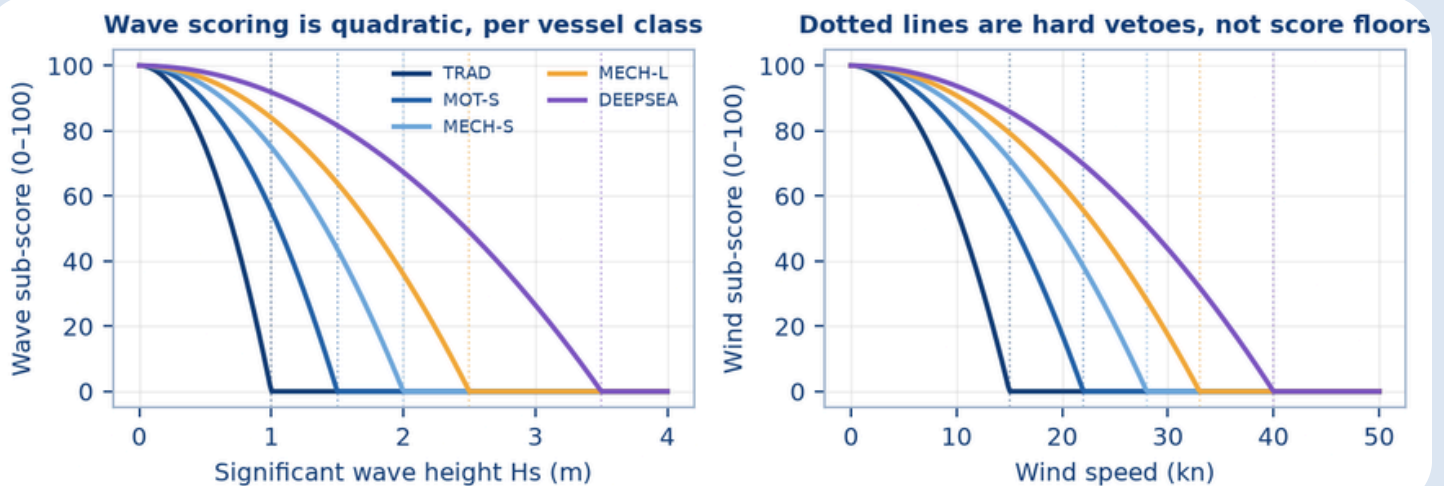


Figure 1: The scoring functions, plotted from the limits the engine actually uses. Dotted verticals are hard vetoes: past them the index is irrelevant because the verdict is already NO-GO.

Five vessel classes, each with cited limits

A 6 m kattumaram and a 30 m deep-sea trawler face the same sea and need opposite answers. ORCA carries five Indian vessel classes with separate wave, wind and visibility limits, each traceable to a published source peer-reviewed work on Indian small-craft safety, and IMD's own small-craft warning bands.

Class	Class	Max Hs	Max wind	Min visibility
Traditional non-motorised	0-7 m	1.0 m	15 kn	2.0 km
Motorised FRP (outboard)	7-10 m	1.5 m	22 kn	2.0 km
Small mechanised	10-15 m	2.0 m	28 kn	1.0 km
Large mechanised	15-24 m	2.5 m	33 kn	1.0 km
Deep-sea	> 24 m	3.5 m	40 kn	0.5 km

22 kn is the bottom of IMD's small-craft warning band; 33 kn is its top. The thresholds are policy, not preference, which is why they are versioned and cited in every response.

4. Boat Safety Index: the physics a wave height alone misses

Significant wave height is a poor predictor of capsizing on its own. A 1.5 m sea with a short period is steep and dangerous; the same height with a long swell is comfortable. ORCA therefore implements the INCOIS Small Vessel Advisory and Forecast Services Boat Safety Index, using the published equations exactly as written.

$$\text{Index} = 0.35 \cdot S_{\text{wave}} + 0.30 \cdot S_{\text{wind}} + 0.15 \cdot S_{\text{visibility}} + 0.20 \cdot S_{\text{instability}}$$

$$S_{\text{wave}} = \max(0, 100 \cdot (1 - (H_s / H_{s_limit})^2))$$

$$S_{\text{wind}} = \max(0, 100 \cdot (1 - (U_{10} / U_{limit})^2))$$

$$S_{\text{vis}} = \min(100, (V / 10 \text{ km}) \cdot 100)$$

$$S_{\text{steep}} = 1 \text{ if } I_{\text{steepness}} > 0.8$$

$$S_{\text{cross}} = 2 \text{ if } I_{\text{crossing}} > 0.65$$

$$S_{\text{rapid}} = 4 \text{ if } Z_{6h} > 0.2$$

Beam criterion: beam < 4 · Hs → capsize-prone when BSI > 0

Aditya, Sandhya, Harikumar & Balakrishnan Nair (2020), Journal of Operational Oceanography. BSI ranges 0-7; the paper treats any non-zero value as dangerous. The weights 1/2/4 are deliberately distinct powers of two so the total is decodable back to which criteria fired.

Three refusals matter as much as the equations. ORCA will not invent directional spread from wave direction alone, will not infer vessel beam from length, and will not map BSI linearly onto its 0-100 index. Where an input is genuinely unavailable, the component is reported as unverifiable rather than estimated.

5. Search and rescue: from a last known position to a search plan

When someone goes into the water, a drift model alone is not enough. It tells you where the object could be; it does not tell you how to look for it. ORCA computes both, and names the rescue centre responsible.

Stage one: Monte Carlo leeway drift

$$\text{drift} = \text{current} + \text{leeway}(\text{wind})$$

$$\text{DWL} = a_{\text{dwl}} \cdot U10 + b \quad (\text{downwind, per particle})$$

$$\text{CWL} = \pm a_{\text{cwl}} \cdot U10 \quad (\text{crosswind, sign drawn per particle})$$

$$\sigma_{\text{current}} = 0.15 \text{ m/s} + 0.25 \cdot |\text{current}|$$

IAMSAR leeway decomposition. Coefficients differ per object class, a person in the water drifts at about 1.1% of wind speed, a canopied liferaft at 4%.

The current-error term was the single most important correction we made. It was originally a flat 0.15 m/s, and the result was a 95% search area of 176-189 km² at four locations whose drift displacements differed by a factor of twenty-one, a constant dressed up as a calculation. Scaling the error with the local flow changed those same four areas to 210-1039 km². A fast, sheared current is exactly where a 0.08° analysis grid smooths away the structure that matters.

Stage two: the IAMSAR search plan

Given the containment area, ORCA computes how to actually search it, using the published sweep-width tables rather than estimates.

$$W = W_{\text{table}}(\text{object, visibility}) \cdot f_{\text{weather}} \cdot f_{\text{fatigue}}$$

$$S = W / C \quad C = 1.0 \text{ for an initial search, } S \geq 0.1 \text{ NM}$$

$$C = (W \cdot V \cdot T) / A$$

$$\text{POD} = 1 - e^{(-C)} \quad \text{POS} = \text{POC} \times \text{POD}$$

W from Table H-19 (Uncorrected Visual Sweep Width, Vessels and Boats) of the USCG Addendum to the National SAR Supplement; f_{weather} from Table H-10, which corresponds to IAMSAR Vol. II Table N-7. Only the POD curve is modelled rather than transcribed, and it is labelled as such.

Why this makes the plan move with the weather

For a person in the water, sweep width is cut to one quarter above 25 knots of wind. The same 200 km² box therefore takes one patrol vessel 9.7 hours to search in a calm and 38.9 hours in a gale because the legs must run four times closer together. A search plan that did not change with conditions would be a picture, not a plan.

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Stage three: who to call, and how long they will take

ORCA carries all 39 Indian maritime rescue centres 3 MRCCs and 36 MRSCs transcribed from Appendix A of the National Maritime Search and Rescue Plan 2022, with their telephone, fax, Inmarsat-C and email. OpenStreetMap was checked first and rejected: a full Overpass query across the Indian box returned 22 features, exactly one of which was a Coast Guard facility. The transit from the nearest centre is routed under Coast Guard limits rather than fishing-boat limits, because a cutter goes to sea in weather that refuses a trawler.

6. Safe passage: routing that refuses rather than reroutes

Route planning runs A* over a lattice built around the direct line, where every node is costed by the same rule engine that answers a point query.

cost(edge) = distance × penalty(risk)

penalty = 1 + 4.0 · shortfall² shortfall = (100 – index)/100

passable = risk.verdict ≠ NO-GO (hard block, not a large cost)

Quadratic, so a mildly worse cell is barely avoided and a nearly-vetoed one is avoided hard. A linear penalty made the planner indifferent between one bad cell and three mediocre ones, which is not how a skipper thinks about a squall.

A vetoed cell is removed from the graph entirely rather than given a high cost. Encoding "never" as a big number is how routers end up sailing through a cyclone to save a day. If no route exists, ORCA says so and explains which cells refused it.

7. The harbour advisory board

A fisheries officer does not want one boat's verdict; they want the fleet's, laid out along their coast. The same rule engine runs at all 60 harbours on the Department of Fisheries PMMSY register, for all five vessel classes, in coastal order, down the west coast from Kutch, round Kanyakumari, up the east coast to the Sundarbans.

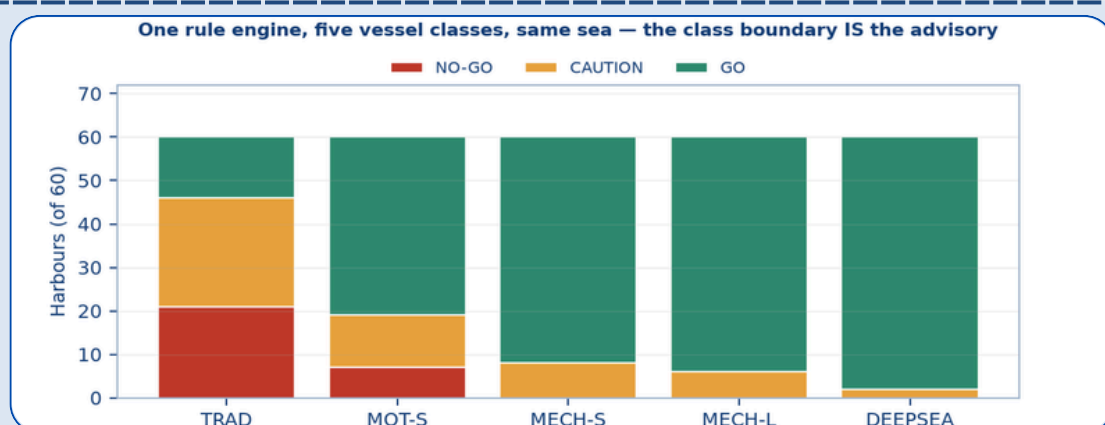


Figure 2: A live board. Identical sea, five classes, and the boundary between them is the advisory: traditional craft are refused at 21 harbours where deep-sea vessels are cleared at 58.

Contiguous runs of the same verdict are collapsed into the sentence an officer can actually broadcast "NO-GO at all 10 harbours from Veraval to Harnai" rather than sixty separate rows. A run never spans both coasts, because a swell on one side of the peninsula says nothing about the other.

8. Speaking the user's language

An advisory nobody can read is not an advisory. ORCA answers in eleven languages spanning the whole Indian coastline, English, Hindi, Gujarati, Marathi, Konkani, Kannada, Malayalam, Tamil, Telugu, Odia and Bengali, by voice as well as text, so a fisherman can ask a spoken question at the quayside and hear the verdict back.

The translation guard, and why it exists

Machine translation is unsafe on operational text in a specific and non-obvious way: it translates the numbers. During testing, "14.7 kn" came back from one provider as "14.7 கிலோ மீட்டர்" , knots silently rendered as kilometres. On a safety threshold, that is the difference between a GO and a capsizing.

So numerals, units and verdict tokens are masked before translation and restored afterwards, never passed to the model as translatable text. Getting that masking right took four iterations, each fixing a real defect: units had to be masked before numerals because the unit pattern is anchored on a preceding digit; the NO-GO token contained a non-breaking hyphen that one provider rewrote; and adjacent mask tokens could merge into an unrecoverable sequence, which is why the index alphabet now excludes the letter X.

The principle underneath

For a person in the water, sweep width is cut to one quarter above 25 knots of wind. The same 200 km² box therefore takes one patrol vessel 9.7 hours to search in a calm and 38.9 hours in a gale because the legs must run four times closer together. A search plan that did not change with conditions would be a picture, not a plan.

9. Data sources

Source	What ORCA takes	Cadence / note
Open-Meteo Marine	Hs, swell, wind-wave, period, direction	Hourly; 20-min per-point cache
Open-Meteo Forecast / ERA5	Wind, gusts, visibility, CAPE, rainfall	Hourly; ERA5 archive from 1940
NASA JPL MUR via NOAA ERDDAP	Satellite SST analysis, 0.01°	Daily; validated against RAMA buoys
ESA CCI via PML ERDDAP	Satellite SST analysis, 0.01°	Monthly composites- stated, not hidden
INCOIS ERDDAP	Regional SST, winds, PFZ inputs	Varies; some archives ended 2023
NOAA PMEL RAMA	Moored-buoy in-situ SST	The only ground truth; ~1 month lag
Marine Regions	EEZ and IMBL treaty boundaries	Static polygons, STRtree-indexed
CelesTrak · AISStream · GFW	Satellite passes, vessel traffic, effort	SGP4 propagation; live AIS
NMSAR Plan 2022 · PMMSY	Rescue centres; fishing harbours	Official registers, transcribed

Table 4 – Nine source families. Every one was probed live before being relied on; those that failed are recorded with the reason rather than quietly dropped.

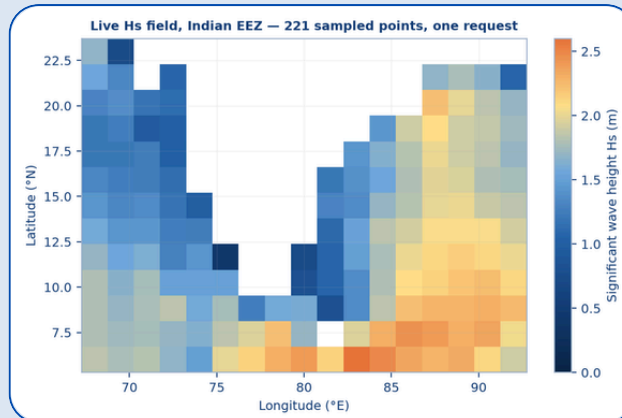


Figure 3: A live significant-wave-height field across the Indian EEZ, 221 points in a single batched request. The white region is the landmass: the wave model's own null response is used as the land mask, which is a better test than any coastline polygon we could ship.

10. The researcher data cube

Researchers lose days reconciling portals, grids and timestamps before any science begins. ORCA's dataset builder takes a plain-language request, variables, a sea area, a date range, and returns one tidy table with one row per day and one column per variable, plus a provenance sheet naming the source, licence and caveat for every column.

Why the variable list is short and true

Thirteen variables are offered, and each earns its place by having been fetched successfully at least once. An earlier version advertised seventeen and delivered two: a column of blanks reads as "measured and absent" when it actually means "never possible", which is worse than not offering the variable at all.

Date	Lat	Lon	SST °C	Hs m	Wind kn	Air °C
2026-08-20	10.5	75.75	26.575	1.8	21.6	28.1
2026-08-21	10.5	75.75	26.624	1.64	25.2	28.3
2026-08-22	10.5	75.75	27.318	1.9	21.8	27.9
2026-08-23	10.5	75.75	27.627	1.7	30	28.4
2026-08-24	10.5	75.75	27.462	1.7	28.3	26.9

Table 5: First five rows of a real 30-day export, built on the deployed server in 8.2 seconds with zero gaps. SST is NASA MUR satellite; Hs is Open-Meteo marine; wind and air temperature are ECMWF ERA5, three providers, one table, each column separately attributed.

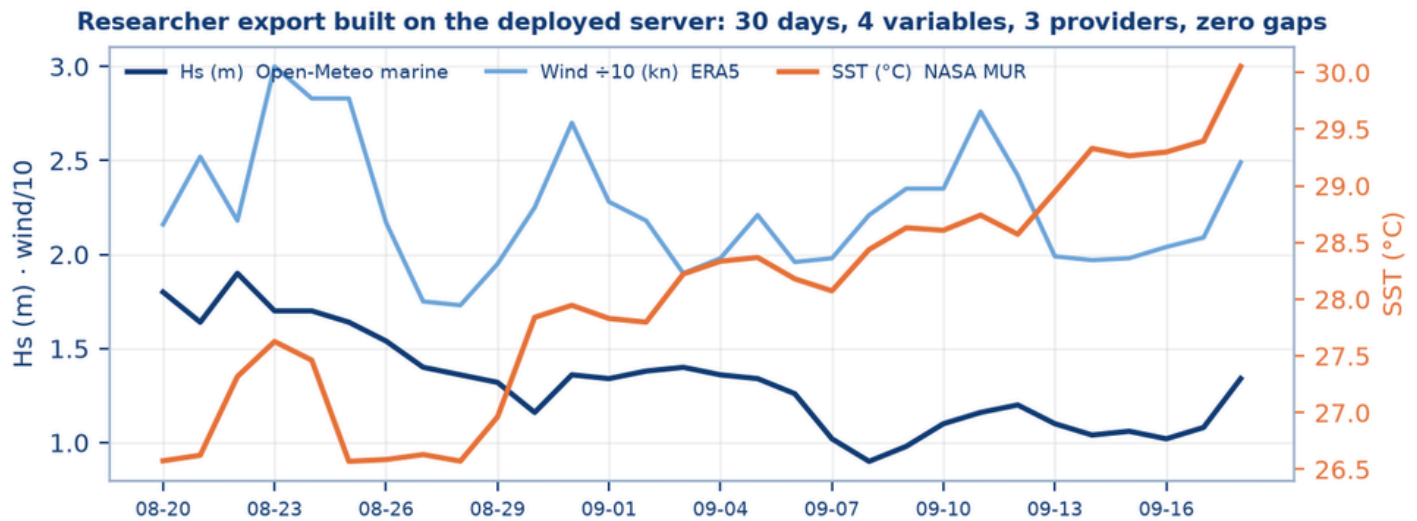


Figure 4: The same export plotted. The mid-window wind peak with rising SST is the kind of structure a researcher would otherwise assemble by hand from three portals.

The cost model that makes it fast

Sources answer different questions and are treated differently. Open-Meteo answers a whole date span in one request, so ninety days of wave height is one call; ERDDAP griddap answers one timestamp, so ninety days of MUR SST is ninety calls. A twenty-day, thirteen-variable build costs 22 upstream requests rather than 260, and the request is costed and refused with the figure before it runs rather than dying at a proxy timeout.

11. Provenance, and the refusal to guess

Nothing reaches the user as a bare number. Every value arrives wrapped in evidence naming its dataset, its age, and one of six provenance states LIVE, CACHED, CURATED, DERIVED, SIMULATED, UNAVAILABLE. A derived value must declare its lineage. An advisory with no citation fails validation at the type level, so an uncited answer is not a representable state.

What this looks like when things go wrong

If the wind feed fails, ORCA does not fill the gap with a plausible number. It returns the verdict it can defend, marks the missing inputs UNVERIFIABLE, lowers its confidence, and tells the user to confirm with their fisheries office or the coastal VHF channel before sailing. A system that is confidently wrong at sea is worse than one that admits the gap.

The same discipline governs what we chose not to build. Storm-surge modelling needs validated bathymetry and expert calibration, so it is absent rather than approximated. Live cyclone tracking is not derived from a best-track archive, because presenting hindsight as a live warning during a cyclone is the worst thing this system could do.

12. Deployment, performance and validation

Measure	Value	How it was obtained
Backend memory	101 MB resident	systems report on the live instance
Harbour board	60 harbours, 2 requests	Batched sampling, order-preserving
30-day 4-variable export	8.2 s, 0 gaps	Timed against the deployed server
Satellite SST vs RAMA buoy	bias -0.05°C , RMSE 0.167°C	30 matched days, station 15n90e
Automated tests	539 passing	pytest, full suite
Frontend end-to-end checks	44 passing	Browser-driven, values compared to API

Table 6: Measured, not estimated. The buoy comparison is the one that matters most: it is the only independent check on the satellite product that every other feature depends on.

Hosting

The frontend is served from a global edge network; the API runs on a dedicated-IP instance behind an automatically renewing TLS certificate, supervised so it survives reboots. The dedicated address matters for a specific reason: free-tier weather APIs count their quota per IP, and on shared hosting that quota is consumed by strangers, which produced rate-limit failures that looked like bugs in our own code.

Verification is part of the engineering, not a phase after it

Several findings in this report came from testing rather than from design. The flat drift error, a router that silently costed passages on wave height alone because an upstream returned HTTP 200 with an error body, and a refresh control that recomputed identical numbers from a cache were all found by measuring behaviour against expectation. Each is fixed, and each has a test that fails if it returns.

13. Honest limits, and what comes next

Limitation	Status
Advisory only, never replaces IMD or INCOIS bulletins	Stated in every response
Coastal-mouth conditions, not offshore forecasts, on the harbour board	Documented in the payload
Buoy ground truth is sparse and roughly a month behind	Reported as coverage, not hidden
MOSDAC adapters written but dormant pending account access	Pending
Cyclone climatology and MPA geofencing designed, not yet built	Sources verified reachable

Table 7: A prototype that overstates its coverage is harder to trust than one that names its edges.